The Wolverine Stack Guide

BPC-157 + TB-500 — Reconstitution, Dosing & Injection

Nurse Rob, RN
nurserobhealth.com · @NurseRobHealth

Educational resource — not medical advice. Consult your physician.

What This Guide Covers

This guide is the clinical reference I wish existed when I started researching peptides. It's built from a licensed RN's perspective — not bro-science, not marketing hype. Just what the research shows and what you need to know to stay safe.

Inside you'll find:

- BPC-157 deep dive — mechanisms, research, dosing protocols
- TB-500 deep dive — what it does, how it differs from BPC-157
- The Wolverine Stack — why these two work better together
- Sterile technique — how to avoid infections (the #1 preventable risk)
- Complete reconstitution & mixing math — exactly how many units for YOUR dose
- Dosing charts for multiple vial sizes and bac water volumes
- Storage, handling, and safety

Whether you're new to peptides or refining your protocol, this guide gives you the math, the methods, and the clinical context to make informed decisions.

BPC-157 Deep Dive

BPC-157 (Body Protection Compound-157) is a synthetic peptide derived from a protein found in human gastric juice. It's one of the most researched peptides for tissue repair and has shown consistent effects across multiple preclinical studies.

What the Research Shows

- Accelerated tendon-to-bone healing by ~40% vs control (J Orthop Res, 2024 rodent study)
- Promoted angiogenesis — the formation of new blood vessels — critical for tissue repair
- Reduced inflammation markers in multiple organ systems
- Showed protective effects on the gastrointestinal lining
- Demonstrated neuroprotective properties in preclinical models

Important Context

Nearly all BPC-157 research is preclinical (animal studies). Human data is limited. Anecdotal reports from the biohacking and sports medicine communities are extensive and generally positive, but these are not controlled trials. The peptide has an excellent safety profile in the available literature.

Standard Research Protocol (Oral)

250-500 mcg taken orally, 1-2 times daily. Oral BPC-157 is typically used for systemic and GI-focused protocols. The acetate salt form (Arg-BPC) is the most commonly available variant. Typical cycles run 4-8 weeks followed by an equal off period.

Standard Research Protocol (Subcutaneous)

250-500 mcg injected subcutaneously, 1-2 times daily. SubQ delivery is preferred for localized healing — injection near the site of injury. The most common approach is 250 mcg 2x/day. Vials are typically 5 mg or 10 mg. See the Reconstitution & Mixing Math section for exact unit calculations.

TB-500 Deep Dive

TB-500 is a synthetic fragment of Thymosin Beta-4, a naturally occurring protein that plays a key role in cell migration, angiogenesis, and tissue regeneration. It's distinct from BPC-157 in both mechanism and dosing schedule.

Key Differences from BPC-157

TB-500 works systemically — it circulates throughout the body promoting actin regulation (cell structure and movement) rather than targeting a specific injury site. Where BPC-157 is local, TB-500 is global. This is why many protocols stack them together: BPC-157 for the injury site, TB-500 for whole-body recovery support.

What the Research Shows

- Promotes cell migration and blood vessel formation (angiogenesis)
- Reduces inflammation and oxidative stress
- Accelerates wound healing in multiple tissue types
- Shown to improve cardiac function post-injury in animal models
- Half-life is approximately 2-3 days — dosed less frequently than BPC-157

Standard Research Protocol

2.5 mg injected subcutaneously, twice per week (5 mg/week total). Some protocols use 5 mg twice weekly (10 mg/week) but 2.5 mg 2x/week is the most widely referenced. Injections are typically spaced 3-4 days apart — e.g., Monday and Thursday. Vials are typically 5 mg or 10 mg.

Because TB-500's effects are systemic, injection location matters less than with BPC-157. Rotate between standard SubQ sites: abdomen, thigh, or upper glute.

The Wolverine Stack

The "Wolverine Stack" is the combination of BPC-157 and TB-500 — named for the character's accelerated healing ability. It pairs a locally-acting peptide (BPC-157) with a systemically-acting one (TB-500) for complementary tissue repair and recovery support.

Peptide	Dose	Frequency	Delivery	Cycle Length
BPC-157	250-500 mcg	1-2x daily	SubQ (near injury) or Oral	4-8 weeks on/off
TB-500	2.5 mg	2x per week	SubQ (abdomen/thigh/glute)	4-8 weeks on/off

Standard Wolverine Stack research protocol. Always consult your physician.

Common Injection Sites

- BPC-157 (SubQ): Near the injury site — shoulder, knee, elbow, or abdomen if systemic
- TB-500 (SubQ): Any standard SubQ site — abdomen (2 inches from navel), outer thigh, upper glute
- Rotate injection sites to prevent tissue irritation
- Clean the site thoroughly with alcohol — see Sterile Technique section

Stacking Logic

These peptides don't compete — they complement. BPC-157 works at the injury through localized angiogenic and anti-inflammatory effects. TB-500 circulates systemically, upregulating actin and promoting cell migration body-wide. Together they cover both the specific repair site and the broader recovery environment.

Sterile Technique

This is the most important section in this guide. Infections from peptide injections are rare but preventable. As an RN, here's exactly what I do and what you should do every single time.

What You Need

- Alcohol prep pads (70% isopropyl alcohol)
- Insulin syringes — 0.3 mL, 0.5 mL, or 1 mL with 31G needle (your standard peptide syringes)
- Clean, flat surface
- Sharps disposal container
- Soap and water — wash hands thoroughly first

The Technique (Every Time)

1. Hand hygiene: Wash hands with soap and water for at least 20 seconds. Dry with a clean towel. This is the #1 infection prevention step and the one most often skipped.

2. Clean the vial top: Swab the rubber stopper of your peptide vial with an alcohol prep pad. Let it air dry completely — don't blow on it, don't wave it around. Air drying = the alcohol actually killing bacteria.

3. Clean the injection site: Use a fresh alcohol prep pad. Swab in a circular motion starting from the center and moving outward. Let it air dry completely — injecting through wet alcohol stings and doesn't sterilize.

4. Use a new syringe every time: Never reuse. Insulin syringes are single-use only. Reusing dulls the needle, increases infection risk, and is never worth saving $0.25.

5. Inject at the correct angle: For SubQ: 45-90 degrees depending on body fat. Pinch the skin, insert smoothly, inject slowly, withdraw. Apply gentle pressure with a clean swab — don't massage the site.

6. Dispose safely: Syringe goes directly into a sharps container. Never recap. Never throw in regular trash.

Reconstitution & Mixing Math

This is where most peptide guides fail. They tell you the dose but not how to calculate it. Here's everything you need — step by step, with tables for every combination.

Step 1: What You Need

Item	Specification	Notes
Bacteriostatic Water	30 mL vial (standard)	Contains 0.9% benzyl alcohol as preservative
Insulin Syringes	0.3 mL, 0.5 mL, or 1 mL, 31G needle	These are what you already use — no special syringes needed
Alcohol Prep Pads	70% isopropyl	One for vial, one for injection site
Peptide Vial	BPC-157: 5 mg or 10 mg TB-500: 5 mg or 10 mg	Lyophilized (freeze-dried) powder

Note: Your standard insulin syringes (the ones you buy in boxes of 100) are all you need. A 3 mL syringe with a 22-25G needle can be used for drawing up bacteriostatic water from the bac water vial if you prefer, but it's entirely optional — you can use multiple draws with your insulin syringe instead.

Step 2: How Much Bac Water to Add

The amount of bacteriostatic water you add determines your **concentration** (mg/mL), which determines how many **units** you draw for your dose. More water = lower concentration = more units per dose. Less water = higher concentration = fewer units per dose.

The standard approach: **1 mL for TB-500 vials** (keeps injection volume small since doses are in milligrams), **2 mL for BPC-157 vials** (gives good precision for microgram dosing). But any volume works — the math is the same.

Step 3: How to Mix (Technique)

1. Swab vial tops (both peptide and bac water) with alcohol. Let dry.
2. Draw air into syringe equal to the amount of bac water you'll add.
3. Inject air into bac water vial first (equalizes pressure).
4. Draw the desired amount of bac water.
5. Inject bac water slowly into the peptide vial, aiming at the glass wall — **not directly at the powder**. Direct impact can damage peptides.
6. Gently swirl (never shake). Let it sit for 2-3 minutes if needed.
7. The solution should be clear. If cloudy or has particles, do not use.
8. Label the vial with the date reconstituted and concentration.

Step 4: The Concentration Formula

This is the only formula you'll ever need:

mg/mL = mg ÷ mL

Example: 10 mg peptide + 2 mL bac water = **5 mg/mL**

To convert to **units on an insulin syringe**:
Units = (desired dose in mcg ÷ 1000) ÷ concentration in mg/mL × 100
Example: Want 250 mcg from a 5 mg/mL vial → (250 ÷ 1000) ÷ 5 × 100 = **5 units**

Dosing Charts

BPC-157 — 5 mg Vial

How many units for your dose, based on bac water added:

Bac Water	Concentration	150 mcg	250 mcg	350 mcg	500 mcg
1.0 mL	5 mg/mL	3 units	5 units	7 units	10 units
1.5 mL	3.33 mg/mL	5 units	8 units	11 units	15 units
2.0 mL	2.5 mg/mL	6 units	10 units	14 units	20 units
2.5 mL	2 mg/mL	8 units	13 units	18 units	25 units

BPC-157 — 10 mg Vial

How many units for your dose, based on bac water added:

Bac Water	Concentration	150 mcg	250 mcg	350 mcg	500 mcg
1.0 mL	10 mg/mL	2 units	3 units	4 units	5 units
2.0 mL	5 mg/mL	3 units	5 units	7 units	10 units
3.0 mL	3.33 mg/mL	5 units	8 units	11 units	15 units
4.0 mL	2.5 mg/mL	6 units	10 units	14 units	20 units

Recommended: 2 mL bac water for 10 mg BPC-157 vials → 5 units = 250 mcg. This is Nurse Rob's standard protocol.

TB-500 — 5 mg Vial

Bac Water	Concentration	2.0 mg	2.5 mg	5.0 mg
1.0 mL	5 mg/mL	40 units	50 units	100 units
1.5 mL	3.33 mg/mL	60 units	75 units	150 units
2.0 mL	2.5 mg/mL	80 units	100 units	200 units

TB-500 — 10 mg Vial

Bac Water	Concentration	2.0 mg	2.5 mg	5.0 mg
1.0 mL	10 mg/mL	20 units	25 units	50 units
2.0 mL	5 mg/mL	40 units	50 units	100 units
3.0 mL	3.33 mg/mL	60 units	75 units	150 units

Nurse Rob's protocol: 10 mg TB-500 vial + 1 mL bac water = 10 mg/mL. 25 units = 2.5 mg per injection, 2x/week. This "double strength" approach keeps injection volume comfortable at 25 units.

Storage & Handling

Proper storage preserves peptide potency and prevents contamination. Here's what you need to know:

State	Temperature	Shelf Life	Notes
Dry powder (unreconstituted)	Freezer (-20°C)	12-24 months	Store in dark. Avoid repeated freeze/thaw cycles.
Reconstituted (in vial)	Refrigerated (2-8°C)	30-60 days	Keep vial upright. Protect from light.
In syringe (pre-loaded)	Refrigerated (2-8°C)	24-48 hours max	Not recommended. Draw fresh when possible.
Room temperature	20-25°C	Days to 1 week	Degrades faster. Refrigerate when possible.

Key Rules

- Never freeze reconstituted peptides — ice crystals damage the peptide structure
- Keep vials in their original packaging or wrap in foil for light protection
- Don't store in the refrigerator door — temperature fluctuates too much
- If traveling, use a small insulated cooler with an ice pack (TSA allows medical coolers)
- When in doubt about sterility or appearance — discard. A $40 vial isn't worth an infection

Next Steps

You now have everything you need to:

- Reconstitute BPC-157 and TB-500 with confidence
- Calculate exact units for any dose, any vial size, any bac water volume
- Follow sterile technique every time
- Store your peptides properly for maximum shelf life
- Understand how the Wolverine Stack works — the clinical why, not just the what

What to Do Now

1. Start a log. Track your doses, injection sites, and how you feel. This data is invaluable — both for you and your provider.

2. Bookmark the formula. mg/mL = mg ÷ mL. It's the only conversion you'll ever need for peptide dosing.

3. Share this guide. If you know someone who's guessing their way through peptide dosing, send them to nurserobhealth.com for the free Wolverine Stack Calculator.

4. Consult a professional. This guide is education, not medical advice. If you're using peptides, have a provider who knows what you're taking and why.

Disclaimer: This guide is for educational purposes only. Nurse Rob, RN provides education, not medical advice. No provider-patient relationship is established by reading this guide. Peptides discussed are for research purposes. Always consult your physician before beginning any new regimen. Do not use peptides to treat, cure, or diagnose any medical condition.

Nurse Rob, RN · nurse@nurserobhealth.com · nurserobhealth.com · @NurseRobHealth